ELNAZ HEIDARI

elnzheidar@gmail.com – SF Bay Area, CA

EDUCATION

Ph.D. in Engineering, Clemson University, SC *Aug 2022 — Aug 2026*
Dissertation: Explainable physics-guided Transformers for flood forecasting (large-scale spatiotemporal forecasting).

M.Sc. in Computer Science, Georgia Institute of Technology, GA *Expected Dec 2026*
Focused on artificial intelligence and machine learning.

TECHNICAL SKILLS

Programming/Scripting	Python, Bash Programming, Go, R&RStudio, C/C++, MATLAB
Web/Cloud Technologies	HPC Clusters, AWS Cloud, Parallel Computation
Machine Learning Libs	Pandas, Numpy, Pytorch, Pydantic, Tensorflow, Keras, Scikit-Learn, CUDA, OpenCV, Scipy, Matplotlib
Deep Learning Networks	Transformers, RNNs, CNNs, Autoencoders, GANs
Reinforcement Learning	PPO, Actor-Critic, Q-Learning, SARSA, Value/Policy Iteration
State Estimation	Kalman Filters, Particle Filters, Bayesian Filtering
Data Services and Tools	NetCDF4, Zarr, HDF5, Xarray, GRIB2, S3 Bucket, AWS CLI, SQL
Version Control and Docs	GitHub Action, ReadtheDocs, HTML5, CSS, LaTeX

PROJECTS

Hydro-Transformers [Transfer Learning, Physics-guided DL, Spatiotemporal Forecasting, CUDA, Pytorch, Shell Scripting, HPC Cluster] **Aug 2022 — Apr 2026**

Developed a large-scale Transformer-based forecasting framework using the Temporal Fusion Transformer (TFT) architecture to model complex spatiotemporal relationships across multiple geographic regions. Trained a unified model on heterogeneous data consisting of dynamic time-series signals and static contextual features, enabling accurate multi-horizon forecasting and uncertainty quantification through quantile regression.

Implemented zero-shot inductive transfer learning to evaluate model performance on unseen regions, demonstrating strong generalization capability. The framework was developed in PyTorch and trained on HPC GPU infrastructure, outperforming NOAA's operational baseline across multiple forecasting horizons.

Reinforcement Learning and Probabilistic State Estimation [PPO, Actor-Critic, GAE, MDPs, Q-Learning, SARSA, Kalman Filter, Particle Filter, PyTorch, Python] **Fall 2026**

Implemented PPO from scratch in PyTorch (Gaussian actor-critic, GAE, clipped surrogate objective) for continuous-control LunarLander, solving it with a 223.7 mean return over 100 evaluation episodes (threshold: 200), with multi-seed ablations of clip range, GAE λ, and learning rate. Modeled traffic-signal control as an MDP with Poisson vehicle arrivals: model-based planning (value/policy iteration) had 35% fewer congestion-threshold violations than Q-learning and 49% fewer than SARSA, and a redesigned reward cut Q-learning violations by 39%. Built a 6-state Kalman filter (matrix operations written from scratch) for multi-target tracking that feeds a safety-aware planner, and a 1,000-particle filter that localizes an orbiting satellite from gravimeter and illumination measurements while also estimating an unknown sun position.

Generative Modeling on MNIST and Fashion-MNIST data utilizing VAEs, GANs and Diffusion Models [PyTorch, PyYAML, Matplotlib, CUDA] **Spring 2025**

Designed, implemented, and benchmarked three cutting-edge generative models including VAEs, GANs, and Diffusion Models on the MNIST and Fashion-MNIST datasets. Through comparative analysis, demonstrated a deep understanding of architectural trade-offs: VAEs consistently produced blurry and grayish, high-diversity samples due to L2 reconstruction loss; GANs generated sharp, high-fidelity images but risked mode collapse and grainy; and Diffusion Models achieved superior results by producing highly sharp, stable, and coherent samples, effectively overcoming the limitations of both prior architectures.

Distributional Semantics and Word Embeddings [PyTorch, GloVe, CBOW, Skip-Gram, Cosine Similarity] **Fall 2025**

Developed and benchmark-validated custom Word2Vec models (CBOW/Skip-Gram) using WikiText-2 with over 2 million token corpus, outperforming pretrained state-of-the-art models with up to 70% analogy accuracy. Quantified model quality via 0.74 cosine similarity on semantic relationships, confirming the successful capture of complex linguistic patterns.

Language Modeling and Text Generation [PyTorch, Numpy, Matplotlib, NLP, RNNs, LSTMs, Transformers, Attention Mechanisms, Gradient Clipping, Positional Encoding, Perplexity Evaluation] **Fall 2025**

Developed and optimized RNN, LSTM, and attention-based sequence models on a custom *Silmarillion* text corpus, reducing test perplexity by approximately 25% through architectural improvements and training techniques including gradient clipping; implemented probabilistic and temperature-based sampling to generate more coherent, stylistically consistent text.

RNNs for Flood Forecasting [Python, Shell, HPC Cluster] **Aug 2022 — Dec 2025**
Developed a cloud-based deep learning framework for flood modeling and uncertainty quantification over 4 regions in SC. Configured GPU-enabled HPC nodes with DL/ML packages and JupyterLab, utilizing NOAA reanalysis weather data and USGS gauging observations for model training. Implemented multiple RNN architectures including LSTM, VRNN and GRU, and benchmarked the models performance against NOAA's National Water Model (NWM V3.0).

SELECTED GRADUATE COURSES

Reinforcement Learning	Fall 2026	**AI for Robotics**	Fall 2026
Machine Learning	Spring 2026	**AI, Ethics, and Society**	Summer 2026
Natural Language Processing	Fall 2025	**Computer Networks**	Summer 2025
Deep Learning	Spring 2025	**Machine Learning for Trading**	Fall 2024
Data Structures and Algorithms	Spring 2024	**ML for Climate Adaptation**	Fall 2023
Risk, Reliability and Uncertainty	Spring 2021		

SELECTED PUBLICATIONS

- Heidari, E., and A. A.Khan. 2026. Probabilistic Sub-Daily Streamflow Forecasting in Data-Scarce Catchments Through a Zero-Shot Inductive Transfer Learning Approach. JAWRA Journal of the American Water Resources Association. https://doi.org/10.1111/1752-1688.70115.
- Heidari, E., Samadi, V., and Khan, A. A. (2025). Leveraging Recurrent Neural Networks for Flood Prediction and Assessment. Hydrology, 12(4), 90.
- Heidari, E., and Khan, A. A. (2026). Interpretable Hourly Flood Forecasting Leveraging Deep Learning Recurrence and Attention Mechanisms. Water Resource Planning and Management.
- Heidari, E., Samadi, V., and Popescu, I. (2023). Flood Modeling Across Mountainous and Coastal Watersheds: Coupling Data-Driven and Catchment Physical Attributes. AGU Fall Meeting.
- Heidari, E., and Samadi, V. (2023). Application of Deep Recurrent Neural Networks for Flood Prediction and Assessment. In EGU General Assembly Conference Abstracts (pp. EGU-16678).